Exploratory Data Analysis and Decision Tree Analysis Report

Basic Information:

Dataset has 32581 number of instances and 12 attributes. Among those 4 are categorical and other are numerical.

About Target variable:

Target variable (loan_status) is highly imbalanced, 78.184% are “0” which means not-defaulted and rest 21.816% are “1” which means defaulted. Any machine learning model would be biased if trained on this data.

Operations on numerical variables:

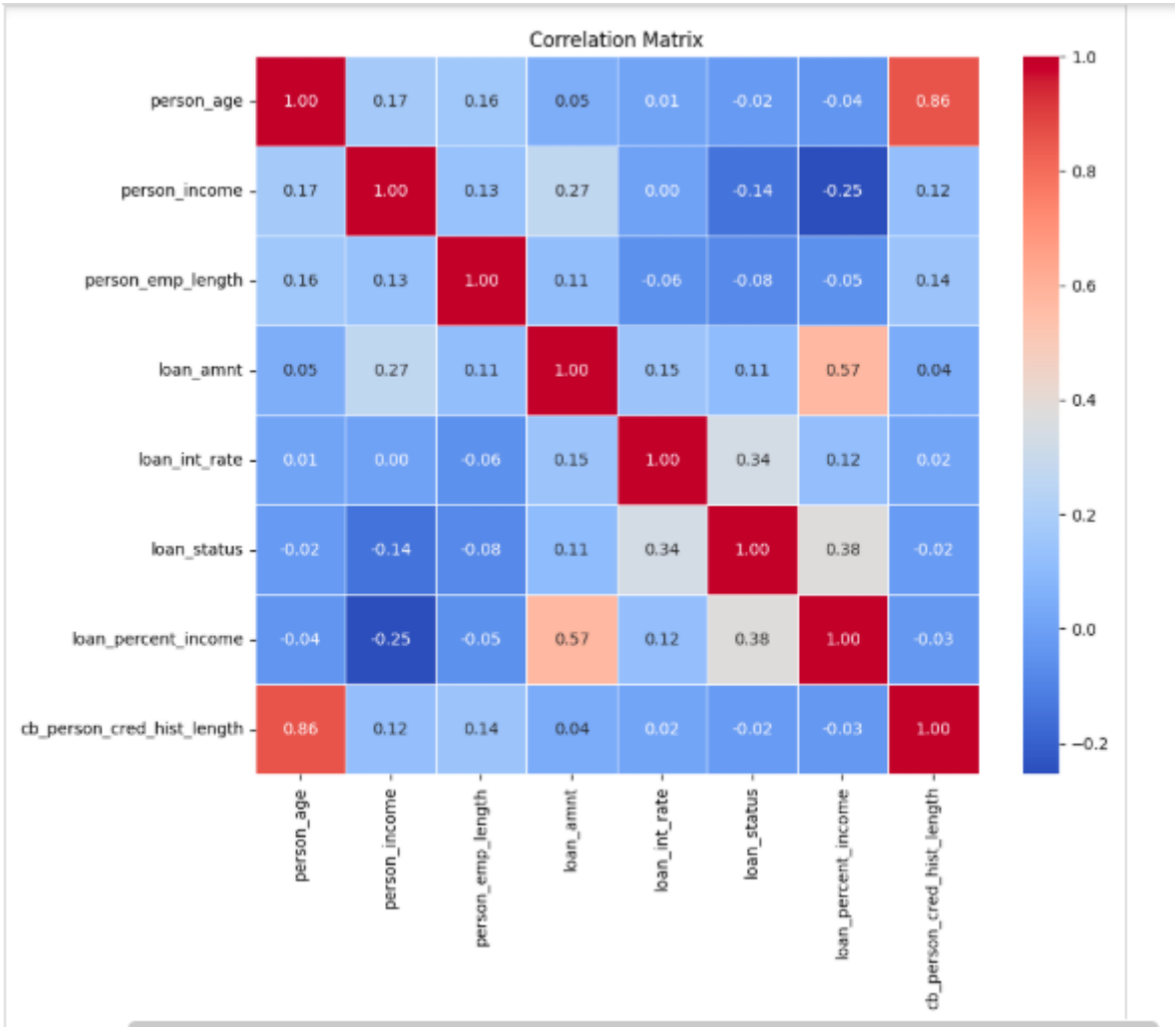

Fig. 1 Correlation Matrix of Numerical variables

From this correlation matrix we can say that Person_age is correlated with person’s credit history length and loan_amount is correlated with loan_percent_income , which is pretty normal.

	0
loan_int_rate	3116
person_emp_length	895
person_income	0
person_age	0
person_home_ownership	0
loan_intent	0
loan_grade	0
loan_amnt	0
loan_status	0
loan_percent_income	0
cb_person_default_on_file	0
cb_person_cred_hist_length	0

dtype: int64

Fig 2. Missing values

loan_int_rate and person_emp_length have 3116 and 895 amount of missing values, respectively.

```
# Display duplicate rows
df_copy.duplicated().sum()
```

```
np.int64(165)
```

Fig 3. Duplicate rows

Dataset have 165 duplicate rows. Those must be removed.

Outlier Detection:

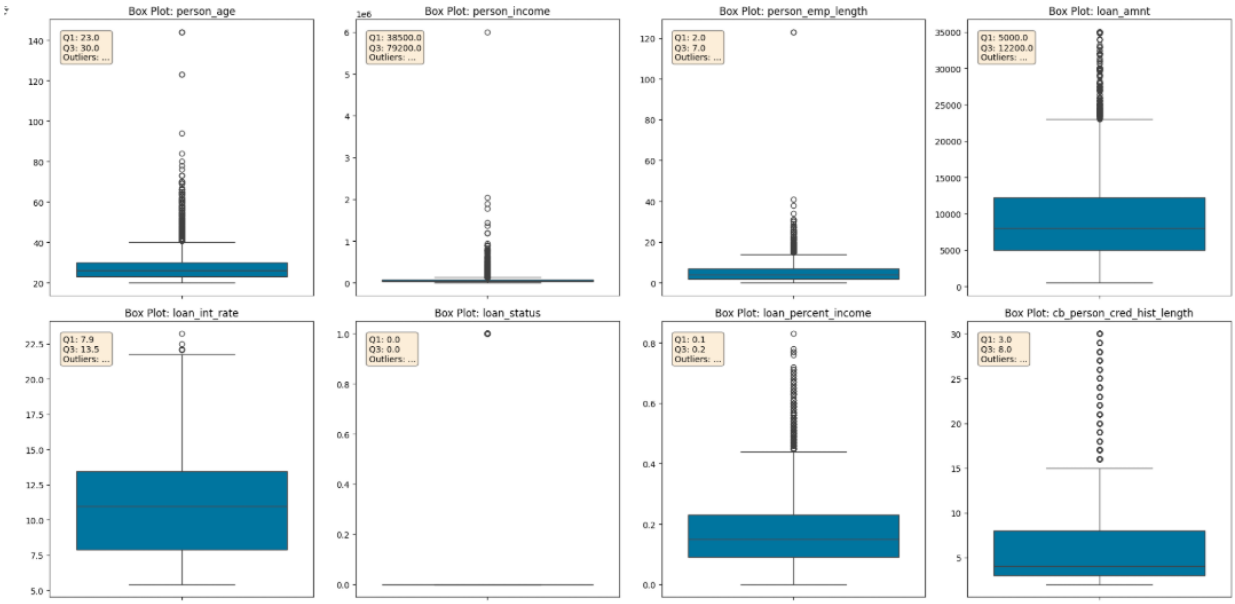

Fig 4. Box plot for outlier finding

This shows person's age (max value 114 years)and person_emp_length (max value is 123 years) which is impossible. According to box plot person_age have outliers from 41 to 114, but that can a real information so we can't remove by IQR. We remove just the impossible values (above 100 years) and for person_emp_length we remove above 50 years.

Distribution, skewness and kurtosis:

Feature	Skewness	Kurtosis	Shape Interpretation
person_age	2.58	18.56	Right-skewed, heavy tails (some very old individuals).
person_income	32.86	2693.27	Extremely right-skewed, extreme outliers (very high-income individuals).
person_emp_length	2.61	43.72	Right-skewed, with some people having much longer employment.

loan_amnt	1.19	1.42	Mild right-skew, close to normal kurtosis.
loan_int_rate	0.21	-0.67	Nearly symmetric, slightly flatter than normal.
loan_status	1.36	-0.14	Slightly right-skewed, nearly normal tails.
loan_percent_income	1.06	1.22	Slight right-skew, close to normal.
cb_person_cred_hist_length	1.66	3.72	Right-skewed, slightly heavy tails.

Table1: Explanation about kurtosis and skewness

Bi – Variate analysis:

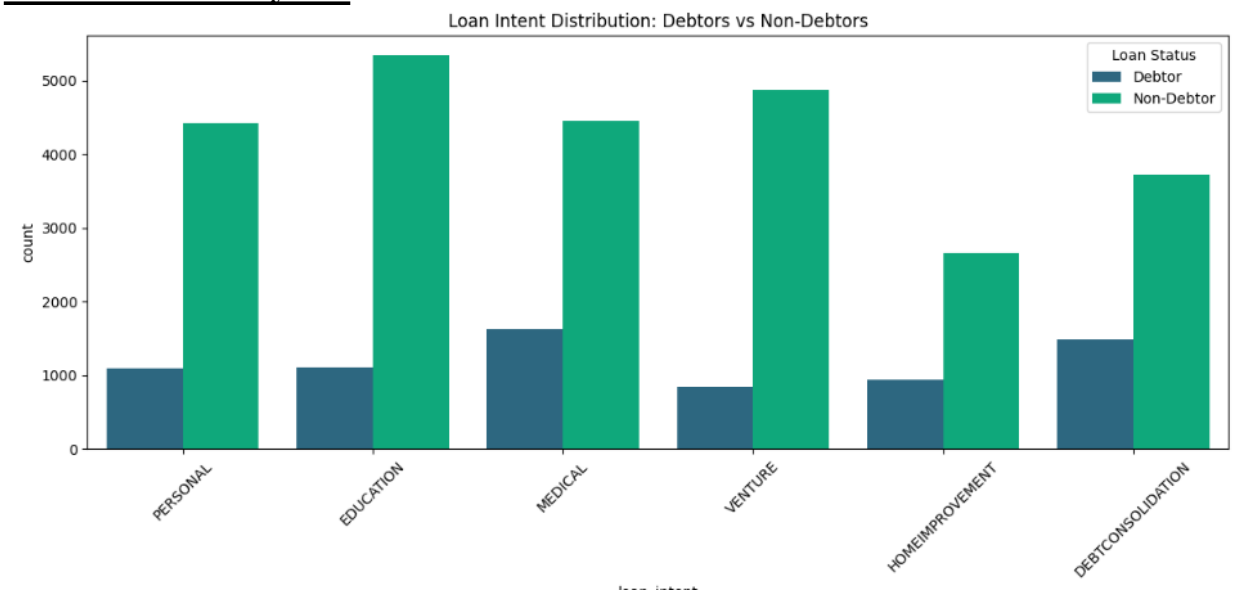

Fig 5. Loan intent distribution vs Loan Status

From this we can say most of loan defaulted are medical loans and most of the loan paid are educational loans.

Decision Tree:

```
DecisionTreeClassifier
DecisionTreeClassifier(class_weight='balanced', criterion='entropy',
                       random_state=42)
```

Fig 6. Decision Tree model parameters

As Target variable is imbalanced we use class_weight = ‘ balanced’ , alternatively we can use SMOTE (Synthetic Minority Over-sampling Technique) to handle class imbalance.

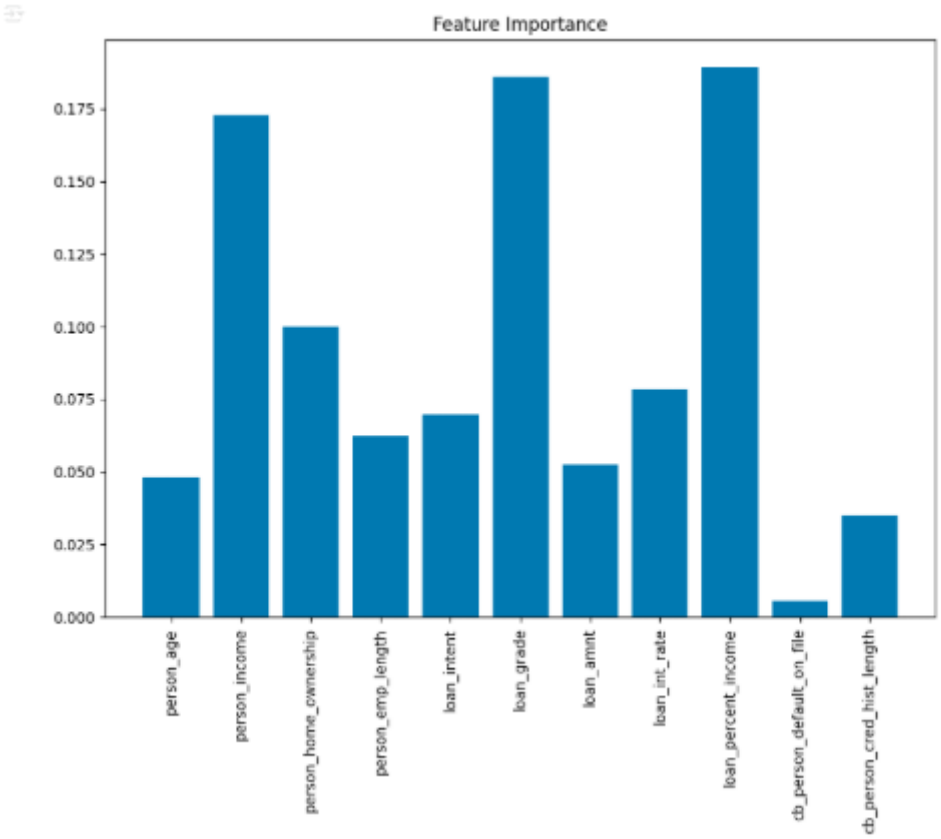

Fig 7. Feature importance

after apply decision tree model we found out that loan_percent_income is the most important feature or in other words we can say that this feature has highest entropy.

```
# depth of the tree
id3.get_depth()
```

```
33
```

```
# getting the score
print(f"Training accuracy : {id3.score(X_train, Y_train)}")
print(f"Test accuracy {id3.score(X_test, Y_test)}")
```

```
Training accuracy : 1.0
Test accuracy 0.8901573588398642
```

Fig 8. Depth of the decision tree and accuracy

From this we can conclude that, this model is over fitted as Training accuracy – Test accuracy > 0.05.

We need to prune the tree.

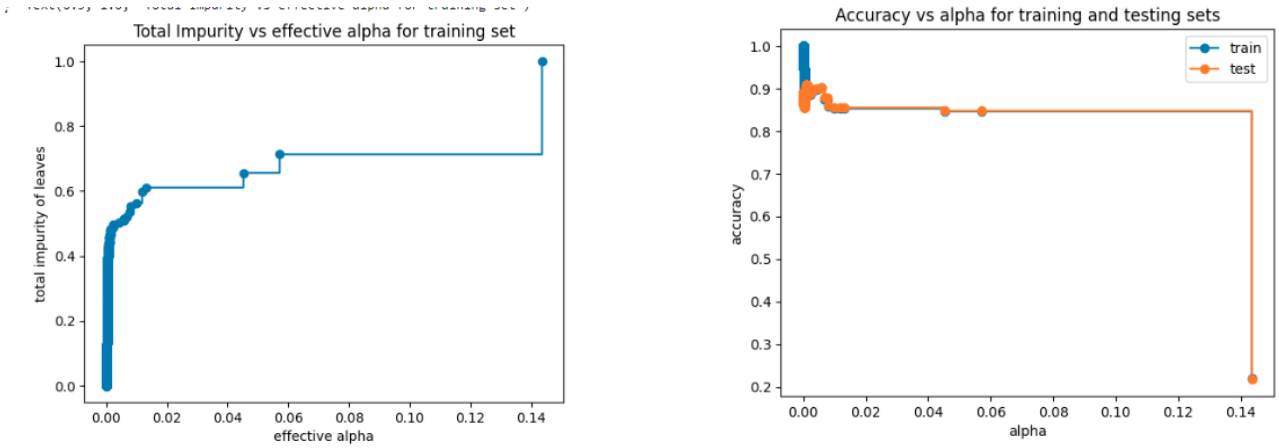

Fig 9. Total impurity and Accuracy vs alpha

We can say 0.001 to 0.005 is the best range for alpha.

```
(0.8969849246231156, 0.8999280057595392)
```

Fig 9. Accuracy after pruning

After pruning the accuracy is most acceptable.